JUDGING SETUP

every judge is also a generator; identities hidden, presentation order randomized

Single-turn

Real-POCQi · 620 questions · 8 models

JUDGE SEES

One clinical question and all eight candidate answers

A B C D E F G H

SCALE

620 questions × 8 judges
4,960 judgments, each covering 8 answers

Multi-turn

MedSP1000 · 200 scenarios · 6 models

JUDGE SEES

One standardized-patient scenario and six clinician transcripts, cut to a fixed number of visible turns

A B C D E F

SCALE

200 scenarios × 6 judges × 4 lengths
4,800 judgments

2 turns | 4 turns | 6 turns | 8 turns

One judgment yields

The judge scores every candidate on accuracy, clinical utility, source quality, verifiability, and completeness, then ranks the full set.

1. **Five axis scores per candidate, 0–5**
2. **One strict ranking of all candidates**

SELF-PREFERENCE METRICS

each compares how the judge treats its own answer with how other judges treat that same answer

Answer Rank Bias — PRIMARY

How many positions higher a judge ranks its own answer than other judges rank it

COMPUTATION

For one question and one judge, say judge C. Example values in boxes.

1. Find **C's own answer** among the candidates. C does not know which one it is. → `answer #5`
2. In **C's ranking**, where did C place that answer? → `2nd of 8`
3. In the **other judges' rankings**, where did they place the same answer, on average? → `4.5th of 8`

↓

ANSWER RANK BIAS

```
step 3 − step 2
= 4.5 − 2 = +2.5 positions
```

Rank 1 is best, so a positive value means C ranks its own answer higher than others do. Averaged over all questions, with judge-by-generator fixed effects to remove presentation-position effects.

↓

OUTPUT

Positions on the candidate list. 0 = no bias; positive = favors own.

Single-turn | **Multi-turn, per length**

Answer Score Bias

How much higher a judge scores its own answer than other judges score it

COMPUTATION

For one question and one judge, say judge C. Example values in boxes.

1. Find **C's own answer** among the candidates. C does not know which one it is. → `answer #5`
2. Average **C's five axis scores** for that answer. → `4.2 / 5`
3. Average the **other judges' five axis scores** for the same answer. → `3.6 / 5`

↓

ANSWER SCORE BIAS

```
step 2 − step 3
= 4.2 − 3.6 = +0.6 points
```

Higher score is better, so a positive value means C scores its own answer higher than others do. Averaged over all questions. Judges calibrate the 0–5 rubric differently, so compare within a judge, not across judges.

↓

OUTPUT

Points on the 0–5 scale. 0 = no bias; positive = favors own.

Single-turn | Multi-turn

Own-Answer Win Rate

How often a judge places its own answer above a competitor

COMPUTATION

For one question and one judge, say judge C. Example values in boxes.

1. Find **C's own answer** among the candidates. C does not know which one it is. → `answer #5`
2. In **C's ranking**, pair that answer with each of the other candidates. → `7 pairs`
3. Count the pairs where C placed **its own answer above** the competitor. → `5 of 7`

↓

OWN-ANSWER WIN RATE

```
step 3 ÷ step 2
= 5 ÷ 7 = 71%
```

Uses only C's own ranking, so it reflects both self-preference and how good the answer actually is. Averaged over all questions.

↓

OUTPUT

Percentage of competitors beaten. 50% = coin flip; 100% = always first.

Single-turn | Multi-turn